

Test solution providers

National Instruments. The 802.11ac WLAN test solution from National Instruments (NI) provides flexibility in testing 802.11ac devices in addition to 802.11a/b/g/n devices. Tarun Gupta, business development manager, telecom and defense, says, "NI PXI-based RF test solutions take advantage of the latest developments in PC buses, multicore processors and FPGAs to deliver testing up to ten times faster than traditional box instruments. All the modulation and demodulation is performed on the host PC rather than the firmware of the instrument, making NI PXI systems extremely flexible to different standards."

Agilent Technologies. Agilent's 802.11ac software allows engineers to view and troubleshoot all 802.11ac modulation formats, from BPSK up to 256QAM, implemented in components and receivers. For even greater flexibility, the software supports all signal bandwidths, including 20, 40, 80 and 160 MHz, and up to 4x4 MIMO.

"Using the 89600B VSA 802.11ac software, engineers gain greater insight into their next-generation 802.11ac WLAN chips and devices, regardless of the 802.11ac format implemented," says Siddiqui, Agilent Technologies.

Rohde & Schwarz. Rohde & Schwarz offers products that can handle the LTE-Advanced and WLAN 802.11ac wide-band communication standards. For instance, the R&S FSW signal and spectrum analyser combines a demodulation bandwidth of 160 MHz with a multistandard radio analysis function. This allows users to simultaneously analyse multiple mobile radio and wireless standards at different frequencies. The R&S SMU200A and R&S SGS100A signal generators complete the offering.

but their number is not enough to warrant a network upgrade.

A similar situation was observed with the 802.11n standard back in 2006. Several pre-standard devices were released by manufacturers. But unfortunately these devices were not completely compatible with the ratified ones, creating a frustrating user experience for consumers.

In September 2011, ABI Research said that established Wi-Fi chipset vendors like Texas Instruments, Qualcomm Atheros and Broadcom will attempt to grow their market share by transitioning as fast as possible. So you can expect these companies to be the first to get the new technology to the market.

Based on the observation of previous events, router and computer manufacturers should be the first to adopt and release 802.11ac standard devices. Quantenna Communications has already launched an 802.11ac gigabit-wireless solution for Wi-Fi routers and consumer electronics at *Consumer Electronics Show (CES)* this year.

Broadcom also launched its first gigabit-speed 802.11ac chips at CES. Rahul Patel, vice president-marketing, WLAN line of business, says, "5G Wi-Fi is expected to penetrate all of Broadcom's product segments. Broadcom expects the first products to hit shelves

in the second half of 2012 starting with access points, followed by consumer electronics devices."

Following routers and PCs, we can expect the huge handset industry to come up with 802.11ac technology on smartphones, tablets and everything in between. This is something to definitely look forward to as the power profile for the new standard is theoretically very promising. Redpine Signals has already announced an ultra-low-power 802.11ac technology for smartphone application processors.

"Our Quali-Fi technology leverages proven low-power innovations spanning over ten years and provides an optimal solution for integration into smartphone application processors and mobile combo chips," says Venkat Mattela, chief executive officer, Redpine Signals.

This will eventually be followed by an assortment of devices which form the rest of the WLAN ecosystem—smart TVs, broadband gateway and Internet TV devices.

What consumers can expect

Once 802.11ac gains momentum, consumers can expect enhanced range and speed of home networks. With Gigabit-wireless speeds, a lot of new usage

scenarios can be expected.

In portable devices like laptops, smartphones and tablets, using 802.11ac Wi-Fi will allow greater reliability for the network while theoretically enhancing the battery life of these devices. Due to faster data transfer, devices will also be able to complete functions quickly and thus improve productivity while using roughly the same power.

802.11ac will enable simultaneous streaming of HD video to multiple clients within the network. This will put the ante in home entertainment systems. Systems that use wireless network-attached storage systems, like Apple's Time Capsule, will be able to leverage the new technology to provide rapid synchronisation or back-up.

Should you upgrade your network?

Since the standard is still in draft, it might take a while for 802.11ac ecosystem devices to gain popularity. Moreover, the devices that first hit the market would be based on the draft version of the standard (currently 1.3) and thus carry a risk. Once the Wi-Fi Alliance ratification is over, we can expect a greater influx of these devices into the market.

At the same time, radios featuring the new standard are backward-compatible with the legacy 802.11a/b/g/n devices. So any legacy device you buy in the meantime will not go wasted.

Heard of 802.11ad?

Seemingly 802.11ac is not the only wireless technology about to hit the market. 802.11ad is another Wi-Fi technology which utilises the 60GHz band to enable data transfer speeds of up to 7 Gbps. On the downside, because this technology uses a very high-frequency band of 60 GHz, it allows short range. A possible use-case would be for transferring HD media wirelessly in a home entertainment system, where this technology could replace conventional HDMI cables. ●

The author is a tech correspondent at EFN